Lessons from Teaching Applied Cryptography in Post-Crisis Lebanon

Real World Crypto 2026

Nadim Kobeissi
https://appliedcryptography.page

Wish you were with us

Suleiman Jaber

Seen here presenting on zero-knowledge proofs

Daniel Dia

Seen here presenting on formal verification of Signal

Flights canceled at the last minute due to war

Lebanon, 2025

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Economic Collapse (since 2019):

- Currency lost **98%** of its value
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2024 Attacks by Israel:

- **4,047** killed, incl. 316 children
- **16,638** injured
- **1.2 million** displaced (out of 5 million total pop.)
- 99,000+ homes, 60+ schools, 60+ hospitals destroyed

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Let's go teach applied cryptography to 55 students!

55 Students with No Prior Cryptography Experience

The Course: CMPS 297AD / 396AI

Applied Cryptography — AUB, Fall 2025

- **55 students**, most with zero prior cryptography experience
- Textbooks: *Joy of Cryptography* (Rosulek) + *Serious Cryptography* (Aumasson)
- Implementation in Go and Rust

Two parts:

1. **Provable Security**: State-separable proofs → ECC & signatures
2. **Real-World Cryptography**: TLS → ZK proofs & post-quantum

There Is a Deployment Problem

Where we concentrate expertise:

- Zurich, Paris, London
- A handful of well-funded labs
- The same conferences

Where cryptography is a lifeline:

- Post-conflict Lebanon
- Communications-blackout zones
- Regions under active surveillance

Students whose relatives were tracked via their phones during the 2024 war understand metadata threats without a lecture.

Deploying expertise where it matters most *is* real-world crypto.

Earning Attention, Not Assuming It

"Diversity is about demographics.
Inclusion is about creating a sense of belonging." — *Dare Obasanjo*

- **Break the intimidation barrier:** hard math is *mostly intimidation*. Show them the notation is scarier than the idea.
- **Create belonging:** guest speakers, posters, community events, a weekly coffee-shop gathering called *The Key Exchange*

The unbelievably quick success was due to creating a sense of belonging in the classroom.

Breaking the Intimidation Barrier

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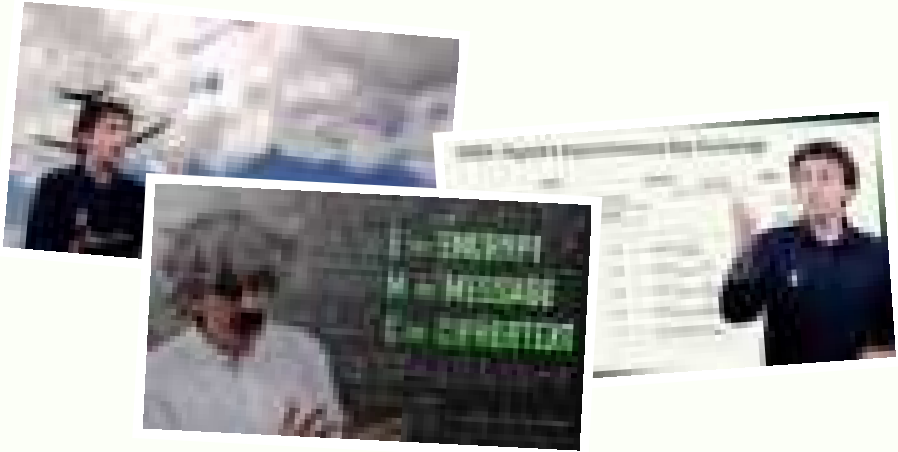

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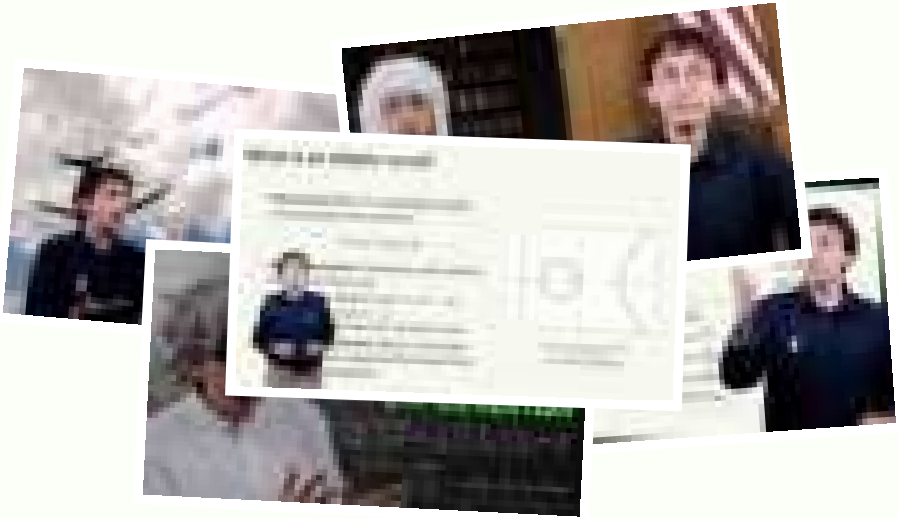

Non-Intimidating Diffie–Hellman

Let p be a safe prime, i.e. $p = 2q + 1$ where q is itself prime. Consider the multiplicative group $\mathbb{G} = (\mathbb{Z}/p\mathbb{Z})^\times$ of order $\varphi(p) = p - 1$. Let g be a generator of the unique cyclic subgroup $\langle g \rangle \leqslant \mathbb{G}$ of prime order q, so that $\langle g \rangle \cong \mathbb{Z}/q\mathbb{Z}$.

The protocol proceeds as follows. Alice samples $a \xleftarrow{\$} \{1, \dots, q-1\}$, computes $A \coloneqq g^a \bmod p$ and transmits A. Bob samples $b \xleftarrow{\$} \{1, \dots, q-1\}$, computes $B \coloneqq g^b \bmod p$ and transmits B. The shared secret is:

$$K = B^a \bmod p = A^b \bmod p = g^{ab} \bmod p$$

Security reduces to the **Decisional Diffie–Hellman** assumption: for all PPT adversaries $\mathcal{A}$,

$$\left|\Pr[\mathcal{A}(g, g^a, g^b, g^{ab}) = 1] - \Pr[\mathcal{A}(g, g^a, g^b, g^c) = 1]\right| \leqslant \mathrm{negl}(\lambda)$$

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Or: Diffie–Hellman with Clocks!

Same concept, except students see *why* primes matter before eventually reaching the notation.

Verifpal: Verification for Everyone

My earlier work on **Verifpal**[a] was built on the same philosophy: protocol analysis shouldn't require a PhD to attempt.

- Replaces the Dolev–Yao π-calculus with an **intuitive language** that reads like a protocol spec
- Students modeled protocols in their *first week* of exposure!
- (Recently rewritten in Rust + new features!)

Lower the barrier, keep (as much as possible) rigor.

But, makes the *opposite* mistake of some formal methods tools.

[a] https://verifpal.com

Establishing Infrastructure: In Numbers

1,500+	slides authored from scratch
8	problem sets, **8** project specifications
350	assignments corrected by hand
100	exams corrected
25	student posters, **15** presentations
12+	invited talks from world experts
5	months, **80+** hour weeks

Going from a standard syllabus to what we built is an enormous bootstrapping effort.
In total, I corrected over 2,300 pages in assignments and exams, alone.
(Hardcore curiosity about how well students are improving is a great motivator!)

Sophia Bou Khuzam on transcript collision attacks

Hala Fatfat making Schnorr protocols and ZK proofs intuitive

Chu Zhang examined vulnerabilities in **Lebanon's own** cellular infrastructure.
Ahmad Abdellatif exposed the economics of zero-day exploits through Zerodium.

Rabab Salim on the Cypherpunk's Manifesto and the Israeli spyware industry

Abd El Kader Kahil on how Rust defeats memory vulnerabilities

15 student presentations across four rounds
Every one of them optional, every one of them packed

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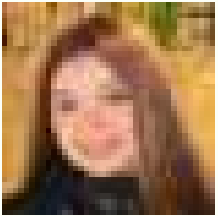
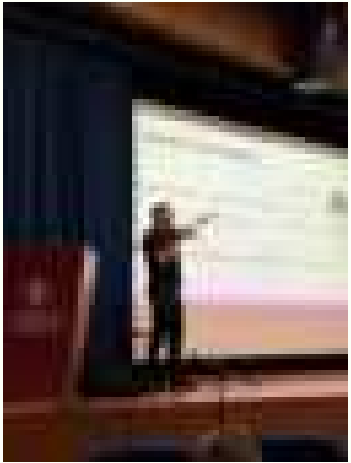
In some cases, students improved on state-separable proofs in their textbooks, producing shorter, more elegant hybrid proofs than those published in the references they were learning from (**Silina Elfil** →).



Guest Speakers (not comprehensive!)

- **JP Aumasson**: flew to Beirut, presented at Bathish Auditorium
- **Kenny Paterson**: end-to-end encrypted cloud storage attacks
- **Rolfe Schmidt**: secure messaging from first principles
- **Daniel J. Bernstein**: designing ECC
- **Dayane Horkos**: Lebanese U. to Paris
- **Vitalik Buterin**: the “cryptography singularity”
- **Nigel Smart**: FHE & threshold crypto



J.P. Aumasson at the American University of Beirut

Poster Session

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After all these distinguished guest lecturers...
here comes our *most persistent* one.

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Israeli drones flew low over campus and rest of Beirut almost daily, often with attached missile pods.
Not very civil! Try teaching ZK proofs under that!

The Results

- **Progression:** Students who found PS1 bewildering achieved near-perfect scores by PS4
- **PS7:** 71% earned A+ or A, 8 perfect scores
- **Midterm:** class average **B+**, designed to international competitive standards
- **Final:** 50 students averaged **82/100** "legitimately not easy"—published for peer assessment[a]

[a] https://appliedcryptography.page/final-exam

Recruitment: Top universities in Europe, North America, and Asia are reaching out to recruit these students

Open Source: A Reproducible Model

https://appliedcryptography.page

- **Everything** published under Creative Commons (CC BY-NC-SA 4.0)
- 1,500+ slides in open-source LaTeX
- Official accompaniment to *Joy of Cryptography* (MIT Press)
- Full starter kits in Go and Rust
- **Battle-tested and peer reviewed!**

Already adopted at:

University of Edinburgh · TU Lübeck · Ruhr University Bochum

If you teach cryptography, you can use all of this tomorrow.

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Wow!

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Oh, my!!

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Is it getting hot in here?

The Uncomfortable Truth

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Teaching with massive real-world impact?
Uh, is that a smart career move?

- Five months of 80-hour weeks. 2,300 pages corrected by hand. Less academic credit than one workshop paper.
- A colleague thinking of leaving academia put it bluntly: the rewards and incentives are going to the wrong places.

We built these incentive structures. We can change them.

Democratize Intellectual Capital

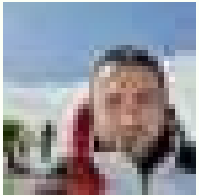
You have an appetite for discussing systemic problems?
Here's one.

The geographical concentration of cryptographic expertise creates dangerous capability gaps precisely where secure systems are most critical.

- Open-source education depends on trust, collaboration, and equality of access: **the exact opposites of gatekeeping, hierarchy, purity, and control.**
- The bottleneck isn't talent or motivation. It's infrastructure and opportunity.
- Fix the incentives so that more people can do this stuff, not just nutty people like me!

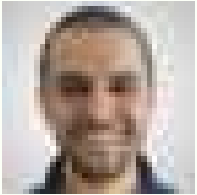
Professors Teaching in Silence

Their teaching doesn't even count towards tenure:



Prof. Mohamed Kobeissi

Prof. Saeed Raheel



Prof. Izzat el Hajj

...and their counterparts at every university you know.

We give each other awards and recognition that don't mean anything on our deathbeds.

Can't we extend some of that recognition to pedagogy?

How About We Just Stop?

How about we all decide to stop allocating resources towards papers we know don't matter?

(Your decade-long quest to get the author of some other paper to respect you doesn't matter, actually.)

We're the ones who make this decision.

We choose what gets funded. We choose what gets published.
We choose what gets recognition.

We can allocate the gold star stickers we ceremonially give one another towards things that are more meaningful.

Announcing: Cedarcrypt

https://cedarcrypt.org

July 13–16, 2026 · AUB Mediterraneo · Paphos, Cyprus

- Summer school + research conference · 2 tracks
- Secure messaging, PQ crypto, ZK proofs, formal verification, privacy technologies
- **Call for submissions open** · Deadline: April 10
- Program Chairs: JP Aumasson & Lucas Meier

Planting a flag: real cryptography work emerges from the Levant.
Be part of our story! **Sponsors welcome.**

If you can do it drone-free, you should.

There are students who need your expertise and faculty who will help you thrive.

- **Course**: `appliedcryptography.page`
- **Cedarcrypt**: `cedarcrypt.org`

Thank you.